

F-test

This test is used to determine whether the variances of two samples are equal or not by dividing them.

F-distribution

The F-distribution is a continuous statistical distribution used to test whether two samples have the same variance.

-The f-distribution has two parameter the numerator degrees of freedom (df1) and the denominator degree of freedom (df2)

$$f\text{-value} = \frac{\text{sample1}/df1}{\text{sample2}/df2}$$

- The independent random variables Samples 1 and 2, have a [chi-square distribution](#).
- The related samples' degrees of freedom are denoted by df1 and df2.

F-test can either be one-tailed or two-tailed depending upon the parameters of the problem.

The F value obtained after conducting an F test is used to perform the [one-way ANOVA \(analysis of variance\)](#) test.

We can use this test when:

- The population is normally distributed.
- The samples are taken at random and are independent samples.

Hypothesis Testing Framework for F-test

For various [hypothesis tests](#) the F test formula is provided as follows:

1. Left Tailed Test:

Null Hypothesis: $H_0 : \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1 : \sigma_1^2 < \sigma_2^2$

Decision-Making Standard: The null hypothesis is to be rejected if the F statistic is less than the F critical value.

2. Right Tailed Test:

Null Hypothesis: $H_0 : \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1 : \sigma_1^2 > \sigma_2^2$

Decision-Making Standard: Dismiss the null hypothesis if the F test statistic is greater than the F test critical value.

3. Two Tailed Test:

Null Hypothesis: $H_0 : \sigma_1^2 = \sigma_2^2$

Alternate Hypothesis: $H_1 : \sigma_1^2 \neq \sigma_2^2$

Decision-Making Standard: When the F test statistic surpasses the F test critical value the null hypothesis is declared invalid.

Example

Example Problem for calculating F-Test

Consider the following example In this we conduct a two-tailed F-Test on the following samples:

	Sample 1	Sample 2
σ	10.47	8.12
n	41	21

Step 1: The statement of the hypothesis is formatted as:

- H_0 : no difference in variances.
- H_1 : difference in variances.

Step 2: Let's calculated the value of the variances in numerator and denominator as F-value= $\frac{\sigma_1^2}{\sigma_2^2}$

- $\sigma_1^2 = (10.47)^2 = 109.63$
- $\sigma_2^2 = (8.12)^2 = 65.99$

$$F_{\text{calc}} = (109.63 / 65.99) = 1.66$$

Step 3: Now, let's calculate the degree of freedom: Degree of freedom = sample - 1 Here we have Sample 1 = $n_1 = 41$ and Sample 2 = $n_2 = 21$

$$\text{Degree of sample 1} = d_1 = (n_1 - 1) = (41 - 1) = 40$$

$$\text{Degree of sample 2} = d_2 = (n_2 - 1) = (21 - 1) = 20$$

Step 4: The usual alpha level of 0.05 is selected because the question does not specify an alpha level. The alpha level should be lowered during the test to half of its starting value. Using $d_1 = 40$ and $d_2 = 20$ in the F-Distribution table. ([link here](#)) and Take $\alpha = 0.05$ as it's not given. Since it is a two-tailed F-test then:

$$\alpha = 0.05/2 = 0.025$$

Step 5: The critical F value is found with alpha at 0.025 using the F table. For (40, 20), the critical value at alpha equal to 0.025 is 2.287. Therefore, $F_{\text{table}} = 2.287$

Step 6: Since $F_{\text{calc}} < F_{\text{table}}$ (**1.66 < 2.287**):

We cannot reject null hypothesis.

$\therefore$ Variance of two populations is similar to each other.

F-Test is the most often used when comparing statistical models that have been fitted to a data set to identify the model that best fits the population.

if $F_{\text{calc}} < F_{\text{table}}$

We cannot reject

if $F_{\text{calc}} > F_{\text{table}}$

We can reject